

Microsoft Azure

A Practical Basics Guide — Cloud Fundamentals with a Data & Analytics Lens

Concepts • Architecture • Diagrams • Examples • Interview Q&A

1. What is Microsoft Azure?

Microsoft Azure is Microsoft's public cloud platform: a huge collection of on-demand computing services — servers, storage, databases, networking, analytics, AI — delivered over the internet and billed on a pay-as-you-go basis. Instead of an organisation buying and maintaining its own physical servers, it "rents" exactly the capacity it needs from one of Microsoft's global datacenters, and can scale that capacity up or down in minutes.

Why this matters to a report analyst: Many of the systems your Power BI reports and Excel workbooks connect to — an Azure SQL Database, an Azure Data Lake, a Power BI/Fabric capacity itself — are Azure resources. Understanding the platform underneath helps you troubleshoot connections, understand refresh/performance issues, and speak the same language as the data engineers who build your pipelines.

1.1 The three cloud service models

Model	What Microsoft manages	What you manage	Example Azure service
IaaS (Infrastructure as a Service)	Physical hardware, virtualization	OS, runtime, data, applications	Azure Virtual Machines
PaaS (Platform as a Service)	Hardware + OS + runtime	Your application code and data	Azure App Service, Azure SQL Database
SaaS (Software as a Service)	Everything, including the application	Just your data and configuration	Microsoft Fabric, Power BI Service, Microsoft 365

1.2 Public, private, and hybrid cloud

- **Public cloud:** Shared infrastructure owned by Microsoft, accessed over the internet — this is Azure itself.
- **Private cloud:** Infrastructure dedicated to a single organisation, often on-premises.
- **Hybrid cloud:** A mix of both, connected together — e.g., **Azure Arc** lets you manage on-premises servers using the same tools as Azure resources.

2. Azure's Global Infrastructure

Azure runs from physical datacenters spread across the world. Understanding how these are organised explains why you're asked to pick a "region" when creating almost anything in Azure.

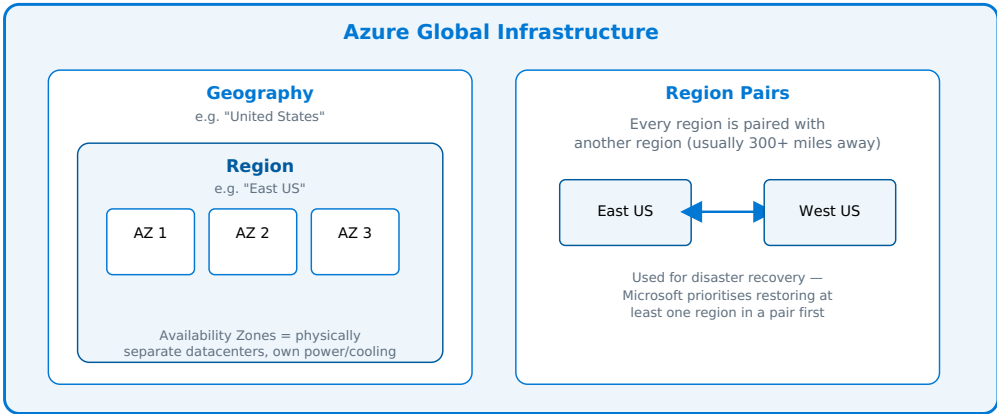


Fig. 1 — Geography → Region → Availability Zones, with regions grouped into disaster-recovery pairs.

2.1 Key terms

- **Region:** A set of datacenters in a specific geographic area (e.g., "East US", "Southeast Asia") — most resources are created "in" a region.
- **Availability Zone (AZ):** A physically separate datacenter within a region, with independent power and cooling — spreading a workload across zones protects against a single datacenter failing.
- **Region pair:** Two regions in the same geography, used by Microsoft for disaster recovery of platform services.

3. Resource Organisation Hierarchy

Everything you create in Azure (a database, a storage account, a virtual machine) is called a **resource**, and every resource must live inside a hierarchy that controls billing, access, and organisation.

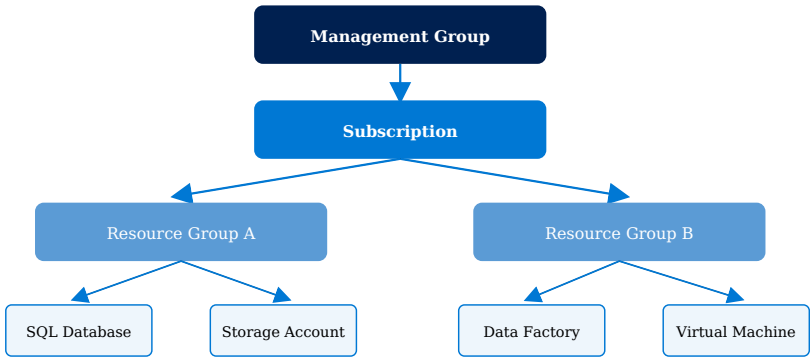


Fig. 2 — Management Group → Subscription → Resource Group → individual Resources.

Level	Purpose
Management Group	Groups multiple subscriptions to apply policy/access consistently across a whole department or company
Subscription	The billing boundary — usage inside it is what generates one invoice; also a hard boundary for some limits/quotas
Resource Group	A logical container for resources that share a lifecycle (e.g., all the pieces of one project) — deleting the group deletes everything inside it
Resource	An individual service instance — a database, storage account, VM, Data Factory, etc.

4. Core Azure Service Categories

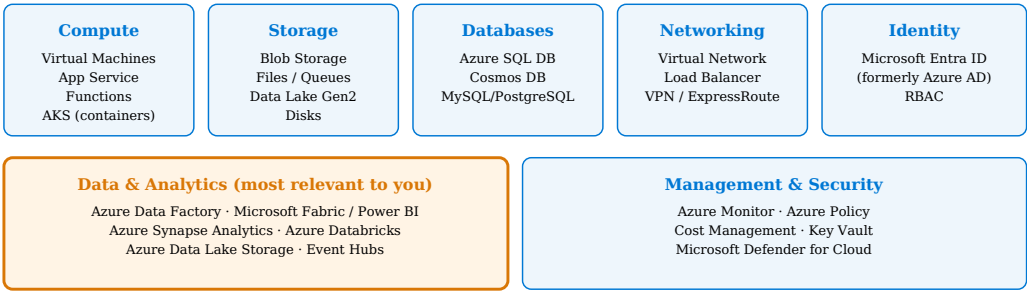


Fig. 3 — Azure's major service categories; Data & Analytics is the one you'll interact with most directly.

4.1 Data & Analytics services relevant to reporting work

Service	What it's for
Azure Data Factory	Cloud ETL/orchestration — moves and transforms data between sources on a schedule
Azure Synapse Analytics	Enterprise-scale analytics combining big-data (Spark) and data-warehousing (SQL pools); Fabric is its evolution into a SaaS product
Azure Data Lake Storage (Gen2)	Massively scalable storage for raw and structured data, built on Blob Storage; often the source Power BI connects to
Azure SQL Database	Fully managed relational database — an extremely common Power BI/Excel data source
Azure Databricks	Managed Apache Spark platform for large-scale data engineering and data science
Microsoft Fabric / Power BI Service	The SaaS analytics and BI layer where your reports and semantic models actually live

5. Identity & Access — Microsoft Entra ID and RBAC

Microsoft Entra ID (formerly Azure Active Directory) is Azure's identity service — it's what checks your username/password (and MFA) when you sign in to Azure, Power BI, or Microsoft 365, and it's the same directory of users and groups used to control who can access what.

5.1 Role-Based Access Control (RBAC)

Azure RBAC decides *what* a signed-in identity is allowed to do, at *which level* of the resource hierarchy (management group, subscription, resource group, or a single resource). Access is granted by assigning a **role** (a bundle of permissions) to a **user, group, or application** at a chosen **scope**.

Built-in Role	What it allows
Owner	Full access, including the ability to grant access to others
Contributor	Full access to manage resources, but cannot grant access to others
Reader	Can view resources, but not change them

Tip: RBAC roles here are conceptually the same idea as Power BI workspace roles (Admin/Member/Contributor/Viewer) — a familiar pattern applied at the whole-platform level.

6. Pricing & Cost Management

- **Pay-as-you-go:** The default model — you're billed only for what you actually consume, by the second/minute/GB depending on the service.
- **Reserved instances / savings plans:** Commit to 1 or 3 years of usage for a discount versus pay-as-you-go.
- **Azure Free Account:** Includes a credit for the first 30 days plus a set of "always free" service tiers, useful for learning.
- **Pricing Calculator:** A public web tool to estimate the monthly cost of a planned set of resources before creating them.
- **Azure Cost Management:** A built-in dashboard to track actual spend, set budgets, and get alerts before you overspend.

7. How This Connects Back to Power BI & Fabric

Microsoft Fabric (covered in the earlier guide) is technically built on Azure infrastructure but sold as one SaaS product — you don't manage the underlying VMs or storage accounts yourself. However, many organisations still keep data in "classic" Azure PaaS services and connect Power BI to them directly:

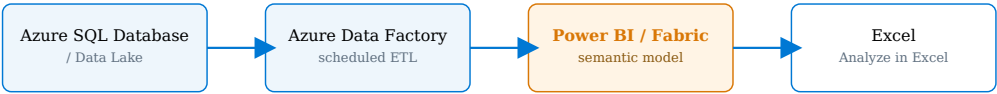


Fig. 4 — A typical "classic Azure" data path into your reporting tools.

8. Quick Glossary

Term	Meaning
Region	A geographic area containing one or more Azure datacenters
Availability Zone	A physically separate datacenter within a region for fault tolerance
Resource Group	A logical container grouping resources that share a lifecycle
Subscription	The billing and quota boundary for a set of resources
Microsoft Entra ID	Azure's identity and access management service (formerly Azure AD)
RBAC	Role-Based Access Control — assigns permissions via roles at a chosen scope
ARM / Bicep	Azure Resource Manager templates / the newer Bicep language — code that defines infrastructure declaratively ("Infrastructure as Code")
SLA	Service Level Agreement — Microsoft's guaranteed uptime percentage for a service

9. Top 10 Interview Questions on Microsoft Azure

1 What is Microsoft Azure, and what are the three cloud service models (IaaS, PaaS, SaaS)?

Azure is Microsoft's public cloud platform, providing on-demand compute, storage, networking and analytics services billed on usage. IaaS gives you raw infrastructure (e.g., Virtual Machines) where you manage the OS upward; PaaS (e.g., Azure SQL Database, App Service) manages the platform for you so you focus on your application and data; SaaS (e.g., Microsoft Fabric, Microsoft 365) is fully managed software where you only manage your data and configuration.

2 What is the difference between an Azure region and an availability zone?

A region is a geographic area containing a cluster of Microsoft datacenters (e.g., "East US"). An availability zone is a physically separate datacenter within that region, with its own power and cooling — deploying across multiple zones protects an application from a single datacenter outage, while regions themselves protect against a much larger, geography-wide event.

3 Explain the Azure resource hierarchy: management groups, subscriptions, and resource groups.

Management groups sit at the top and let policies/access be applied across many subscriptions at once. A subscription is the billing and quota boundary. A resource group is a logical container beneath a subscription that groups related resources sharing the same lifecycle — deleting the resource group deletes everything inside it. Individual resources (a database, storage account, etc.) sit inside a resource group.

4 What is Microsoft Entra ID, and how does RBAC control access in Azure?

Microsoft Entra ID (formerly Azure Active Directory) is Azure's identity service, handling sign-in and holding the directory of users, groups, and applications. Role-Based Access Control (RBAC) then decides what an identity can do by assigning a role — a bundle of permissions like Owner, Contributor, or Reader — to that identity at a chosen scope (management group, subscription, resource group, or single resource).

5 What's the difference between Azure Blob Storage and Azure Data Lake Storage Gen2?

Both are built on the same underlying blob storage technology and are highly scalable and cost-effective for large volumes of unstructured data. Data Lake Storage Gen2 adds a hierarchical namespace on top (true folders and directories rather than a flat namespace) plus fine-grained POSIX-style permissions, which makes it better suited to big-data analytics workloads like those feeding a Fabric Lakehouse.

6 What is Azure Data Factory, and how does it relate to Dataflows Gen2 in Fabric?

Azure Data Factory is Azure's cloud ETL and orchestration service — it moves and transforms data between sources on a schedule using pipelines. Fabric's Data Factory experience (including Dataflows Gen2) brings this same pipeline/dataflow concept into the SaaS Fabric environment, landing its output directly into OneLake instead of a separate Azure storage account.

7 What is the difference between horizontal and vertical scaling in Azure?

Vertical scaling ("scaling up/down") changes the size of a single resource — e.g., moving a VM or database to a bigger SKU with more CPU/RAM. Horizontal scaling ("scaling out/in") adds or removes instances of a resource running in parallel — e.g., adding more VM instances behind a load balancer. Horizontal scaling generally offers better resilience since it doesn't depend on one larger machine.

8 What is Infrastructure as Code, and what tools does Azure provide for it?

Infrastructure as Code (IaC) means defining and provisioning infrastructure (VMs, networks, databases) through declarative configuration files instead of manual clicks in a portal, making deployments repeatable and version- controlled. Azure's native tools are ARM (Azure Resource Manager) templates, written in JSON, and Bicep, a newer, simpler language that compiles down to ARM templates.

9 How is Azure billed, and what tools help control cost?

Azure is primarily pay-as-you-go, billed per second/minute/GB depending on the service, aggregated to the subscription for one invoice. Reserved instances or savings plans offer discounts for 1- or 3-year usage commitments. For cost control, the Pricing Calculator estimates cost before deployment, and Azure Cost Management tracks actual spend, supports budgets, and can send alerts when spending approaches a threshold.

10 How does Microsoft Fabric relate to Azure as a whole?

Fabric runs on top of Azure's global infrastructure but is delivered as a single SaaS product rather than a collection of services you provision yourself — there are no VMs, storage accounts, or clusters to manage directly. It replaces what would previously have been separately configured Azure services (Data Factory, Synapse, Data Lake Storage) with one unified experience, one security model, and one capacity, while still ultimately running on Azure's datacenters.